

README

Isaac Sim on AWS EC2 — Complete Setup & Operations Guide

Hospital Simulation: Operating Theatre (OT) & Patient Room

Version 1.0 | June 2026 | Logicalminds

1. What Is This Document?

This guide is written for anyone — including people with no prior cloud or simulation experience — who needs to spin up the Isaac Sim hospital environment on an AWS EC2 server, connect to it remotely, and perform basic tasks inside the simulation. Follow each step in order and you will have a fully running 3D hospital simulation on your screen.

☐ Read this entire document once before starting so you know what to expect.

2. What You Need Before You Start

2.1 Accounts & Access

- An AWS account with access to EC2 (Asia Pacific — Mumbai region ap-south-1)
- Login credentials for the AWS Console: <https://console.aws.amazon.com>
- A web browser (Firefox or Chrome recommended)
- Your EC2 key pair file (.pem) if connecting via SSH

2.2 Your EC2 Instance Details

Item	Details
Instance Name	Isaac-sim-server
Instance ID	I-07be57fe3a50e6166
Region	Asia Pacific — Mumbai (ap-south-1)
OS	Ubuntu (Linux)

DCV Port	8443
IAM Role	EC2-SSM-Role

3. Step 1 — Start Your EC2 Instance

The EC2 instance is your cloud server. Think of it as a powerful computer sitting in Amazon's data center that you turn on and off as needed.

3.1 How to Start the Instance

1. Open your browser and go to: <https://console.aws.amazon.com>
 2. Log in with your AWS credentials
 3. In the top search bar, type EC2 and click on it
 4. In the left menu, click Instances
 5. Find Isaac-sim-server in the list
 6. Check the box next to it
 7. Click the Instance state button at the top
 8. Click Start instance
 9. Wait 2 minutes for the instance to fully start
- ✓ The status will change from Stopped → Pending → Running. Wait until it says Running.

3.2 Get the IP Address

Every time the instance starts, it gets a new public IP address. You need this to connect.

10. Click on the Instance ID (blue link) of your instance
 11. Scroll down to find Public IPv4 address
 12. Copy this IP address — you will use it in the next steps
- The IP address changes every time you restart the instance. Always check the new IP before connecting.

4. Step 2 — Connect to the Server Using Amazon DCV

Amazon DCV (Desktop Cloud Visualization) lets you see and control the remote server's desktop through your web browser — just like sitting in front of the computer in Amazon's data center.

4.1 Create a DCV Session (First Time or After Restart)

Before you can connect via DCV, a session must be created on the server. This is required every time the server restarts.

13. In the AWS Console, go to EC2 → Instances
14. Select your instance → click Connect button
15. Click the EC2 Instance Connect tab
16. Click Connect — a terminal window opens in your browser
17. In the terminal, type this command and press Enter:

```
dcv create-session mysession --owner ubuntu
```

18. Verify the session is running:

```
dcv list-sessions
```

✓ You should see: Session 'mysession' — Status: Active. Session is now ready.

4.2 Open DCV in Your Browser

19. Open a new browser tab
 20. Type in the address bar (replace with your current IP):
- ```
https://YOUR-INSTANCE-IP:8443/?sessionId=mysession
```
21. Your browser may show a security warning — click Advanced → Proceed anyway
  22. Enter your login credentials when prompted
  23. You should now see the remote desktop of your EC2 server

□ If you see 'No session is available', go back to Step 4.1 and create the session again.

---

## 5. Step 3 — Launch Isaac Sim

Isaac Sim is NVIDIA's simulation platform where the hospital environment lives. Once you are connected via DCV, follow these steps to open it.

### 5.1 Open a Terminal on the Remote Desktop

24. Inside the DCV remote desktop, right-click on the desktop
25. Click Open Terminal (or find the terminal icon in the taskbar)

### 5.2 Launch Isaac Sim

Type this command in the terminal to start Isaac Sim:

```
cd ~/.local/share/ov/pkg/isaac-sim-*/
./isaac-sim.sh
```

Isaac Sim will take 3 to 5 minutes to load for the first time. You will see a splash screen and loading bar.

✓ If Isaac Sim is already pinned to the taskbar or desktop, you can simply double-click its icon to launch it.

□ Do not close the terminal while Isaac Sim is loading. Keep it open in the background.

---

## 6. Step 4 — Open the Hospital Environment

The hospital environment includes a fully modelled Operating Theatre (OT) and Patient Room built in NVIDIA Isaac Omniverse. Here is how to open it.

26. Once Isaac Sim is open, go to the top menu: File → Open

27. Navigate to the hospital environment file location:

Isaac Sim → Environments → IsaacLab → (your hospital USD file)

28. Click on the hospital environment file (.usd)

29. Click Open

30. Wait for the scene to load — all assets including the OT and Patient Room will appear

31. Press Play (the triangle button) at the top of the viewport to start the simulation

✓ You should now see the 3D hospital with the Operating Theatre and Patient Room rendered in real time.

□ If any assets appear missing or pink, it means the file path is broken. Contact the technical team.

## 7. Step 5 — Basic Things You Can Do in the Simulation

### 7.1 Navigate the 3D View

| Action             | How to Do It                          |
|--------------------|---------------------------------------|
| Rotate the view    | Hold right mouse button and drag      |
| Move around (pan)  | Hold middle mouse button and drag     |
| Zoom in / out      | Scroll the mouse wheel                |
| Focus on an object | Click the object, then press F        |
| Reset view         | Press Numpad 0 or go to View → Camera |

### 7.2 Select and Inspect Objects

32. Click on any object in the scene (e.g. the operating table, monitor, bed)

33. The object will highlight and its properties will appear on the right panel

34. You can see the object name, position, size, and material settings

### 7.3 Start and Stop the Simulation

- Press the Play button (triangle) to start the simulation running
- Press the Pause button to pause it
- Press the Stop button (square) to reset the simulation to its starting state

### 7.4 Switch Between Operating Theatre and Patient Room

35. Look at the Stage panel on the right side of the screen
36. You will see a list of all objects in the scene
37. Click on OperatingTheatre or PatientRoom to select that section
38. Press F to zoom into that area in the viewport

---

## 8. Step 6 — Connect via SSM Session Manager (Advanced)

SSM Session Manager is another way to access the server's terminal directly from the AWS Console without needing SSH keys. This is useful for running commands and troubleshooting.

### 8.1 How to Connect

39. Go to AWS Console → EC2 → Instances
  40. Select Isaac-sim-server
  41. Click Connect
  42. Click the SSM Session Manager tab
  43. Click Connect
  44. A terminal opens — you are now inside the server
- ✓ SSM Session Manager requires the IAM role EC2-SSM-Role and the SSM agent to be running on the server.

### 8.2 If SSM Shows Offline

If the SSM ping status shows Offline, run these commands via EC2 Instance Connect:

```
sudo snap restart amazon-ssm-agent
sudo snap services amazon-ssm-agent
```

- ✓ Status should show: active and enabled. Refresh the SSM page after 1-2 minutes.

---

## 9. Troubleshooting — Common Issues & Fixes

| Problem                             | Cause                                           | Fix                                                                               |
|-------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------|
| DCV shows 'No session available'    | Session not created or expired                  | Run: <code>dcv create-session mysession --owner ubuntu</code>                     |
| SSM Agent Offline                   | IAM role not attached or agent stopped          | Restart agent: <code>sudo snap restart amazon-ssm-agent</code>                    |
| SSM role error (status 400)         | Trust policy or permissions missing in IAM role | Go to IAM → EC2-SSM-Role → fix trust policy & attach AmazonSSMManagedInstanceCore |
| systemctl command not found for SSM | Ubuntu uses Snap, not systemctl for SSM         | Use: <code>sudo snap restart amazon-ssm-agent</code>                              |

|                                         |                                              |                                                       |
|-----------------------------------------|----------------------------------------------|-------------------------------------------------------|
| <b>New IP address after restart</b>     | EC2 assigns new public IP on every start     | Always check EC2 console for new IP before connecting |
| <b>Isaac Sim takes too long to load</b> | Large scene files, first-time shader compile | Wait 5-10 minutes on first launch — normal behaviour  |
| <b>Pink or missing objects in scene</b> | Broken USD file path                         | Re-open the file or contact the technical team        |

## 10. How to Properly Shut Down

Always shut down properly to avoid losing work and unnecessary AWS costs.

45. Inside Isaac Sim — go to File → Save to save your scene
46. Close Isaac Sim
47. Go to AWS Console → EC2 → Instances
48. Select Isaac-sim-server
49. Click Instance state → Stop instance
50. Wait for status to show Stopped

❑ Stopping the instance stops AWS charges. Do NOT terminate the instance — that deletes everything permanently.

## 11. Quick Reference Card

| Task                           | Command / Location                                                                                |
|--------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Start EC2 instance</b>      | AWS Console → EC2 → Instances → Start                                                             |
| <b>Connect via DCV browser</b> | <a href="https://YOUR-IP:8443/?sessionId=mysession">https://YOUR-IP:8443/?sessionId=mysession</a> |
| <b>Create DCV session</b>      | <code>dcv create-session mysession --owner ubuntu</code>                                          |
| <b>List DCV sessions</b>       | <code>dcv list-sessions</code>                                                                    |
| <b>Launch Isaac Sim</b>        | <code>./isaac-sim.sh</code> (in Isaac Sim install folder)                                         |
| <b>Restart SSM agent</b>       | <code>sudo snap restart amazon-ssm-agent</code>                                                   |
| <b>Check SSM agent status</b>  | <code>sudo snap services amazon-ssm-agent</code>                                                  |
| <b>Stop EC2 instance</b>       | AWS Console → EC2 → Instances → Stop                                                              |